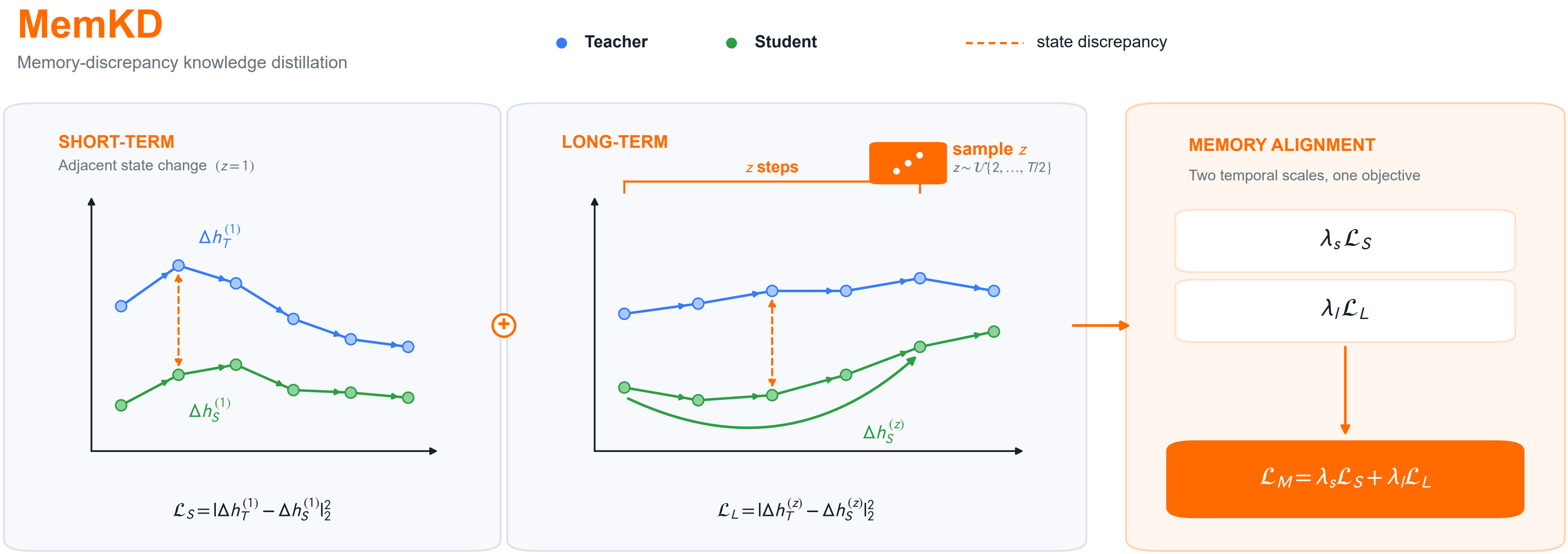

MemKD
Memory-discrepancy knowledge distillation
Teacher
Student
state discrepancy
SHORT-TERM
Adjacent state change ($z=1$)
$\Delta h_T^{(1)}$
$\Delta h_S^{(1)}$
$\mathcal{L}_S = |\Delta h_T^{(1)} - \Delta h_S^{(1)}|_2^2$
LONG-TERM
z steps
sample z
$z \sim \mathcal{U}\{2, \ldots, T/2\}$
$\Delta h_S^{(z)}$
$\mathcal{L}_L = |\Delta h_T^{(z)} - \Delta h_S^{(z)}|_2^2$
MEMORY ALIGNMENT
Two temporal scales, one objective
$\lambda_s \mathcal{L}_S$
$\lambda_l \mathcal{L}_L$
$\mathcal{L}_M = \lambda_s \mathcal{L}_S + \lambda_l \mathcal{L}_L$